

Bearing fault diagnosis from vibration spectra

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September 2026 github.com/Mmaotttt/bearings-adventure

Abstract

I worked through a complete bearing-diagnosis chain on the public Case Western Reserve University (CWRU) dataset, from time-domain waveforms to a decision rule I can defend, and set a supervised classifier against it as a control. A direct spectrum does not locate the fault frequency; the envelope spectrum does. The outer-race and inner-race records reached peak-to-background ratios of **635** and **237** at their own predicted frequencies, against a healthy-record floor of **9.7**, with measured frequency errors of **+0.4%** and **-0.25%**. The resonance band was not chosen by eye: a harmonic-comb score that is blind to which fault is present searched 55 candidate bands, and the decision threshold was calibrated as the highest score the healthy record reaches under the identical search, **11.1**. Of nine fault records, all six race faults were diagnosed correctly and **one false positive occurred** — the 0.021 in ball fault, called inner race. **The ball fault was not diagnosed at any diameter.** The classifier reached **96.3%** when grouped by file, but fell to **33.1%** on time-domain features when the held-out fault diameter was unseen, against a three-class chance level of **33.3%** — not a weak effect, no effect. The pipeline also carries an external check: the kurtosis of all **36** drive-end fault records agrees with the published benchmark study [3] to within **0.3%**.

1. Motivation

In July 2026 I spent two weeks (13 to 24 July) as a robot test engineer in the Technical Department of Wuhan Jingtian Electric Co., Ltd. (武汉京天电器有限公司), a robotics firm trading as Jingtian Robotics.

The project I worked on there was getting a humanoid robot to write calligraphy with a brush on a table, by teach-and-playback: the arm's torque was released, a person held its hand and wrote the character once, the arm's joint angles were recorded, and the robot replayed them. The character we wrote by hand came out normally; the replayed one often did not — the same stroke too thick, or too thin, and sometimes the brush never reached the paper. The weight of a stroke had never been measured, because there was no force sensor on the machine. How hard to press is something a hand corrects continuously by feel, and what entered the record was the joint angles after the correction, not the correcting. So we removed the uncertainty from the environment one source at a time: the paper taped to a table at a fixed height, the brush tip worked back to the same condition before each attempt, the stance watched so that it did not drift, since the waist and leg angles were not recorded, and the gravity compensation, which we wrote ourselves, fixed to that one working height because the writing only ever happened there. Seven or eight takes per character, and the best one kept. It was a reasonable call: a demonstration rather than an industrial task, the time was short, and the crude route finished the job, which it did. But nothing in it was a measurement — the result was bought by making the world repeatable. What stayed with me was not that I could not read some signal, but that it had not occurred to me to want one.

There is no Signals and Systems course in my curriculum. I did this project alongside self-study to close that gap. I had no equipment and no real machine to take apart, and that one constraint decided most of what followed. The data had to be public; and since I was teaching myself, with nobody beside me to say whether an answer was wrong, it had to be a dataset other people had published results on, so that I could at least hold my own numbers against theirs (section 4.0). The same reasoning chose rolling bearings: the fault frequencies follow from the geometry, which means that before touching the data there is already an answer the data ought to give, rather than only an answer the data will supply.

Three sentences of background. The impact produced when a rolling element strikes a defect is broadband; its energy goes into the structural resonance it excites, and the fault frequency is carried on that resonance band as amplitude modulation, which is why envelope analysis is the standard way to recover it [1]. For choosing the band itself, the usual criterion in the literature is spectral kurtosis and its fast implementation, the kurtogram [2]; section 3.4 explains why I used a different criterion on this data. The CWRU dataset already has a published assessment [3], in which three established diagnostic techniques were applied across the whole dataset and every record was graded by how diagnosable it proved; I compare against it in sections 4.0, 4.1, 4.5, 4.6 and 5.2.

2. Data

2.1 Source and structure

The data come from the CWRU Bearing Data Center, drive-end accelerometer channel (DE). The test bearing is an SKF 6205-2RS JEM deep-groove ball bearing. Faults are single points introduced by electro-discharge machining at 0.007, 0.014 and 0.021 in diameter. The catalogue is in `src/dataset.py`, the records are fetched by `src/download_data.py`, and `src/verify_data.py` checks them.

Table 1. The records used.

Records used	40
Structure	healthy + 3 fault locations $\times$ 3 diameters, each at 4 motor loads
Sampling rate	12 kHz (fault records); 48 kHz for the baselines, decimated by 4
Record length	$\sim$ 10 s, except file 97 at 5.08 s
Motor loads	0 / 1 / 2 / 3 hp at approximately 1797 / 1772 / 1750 / 1730 rpm
Segments for classification	2048 samples (0.171 s), non-overlapping, 2331 rows

The segment count is 2331 rather than $40 \times 59 = 2360$ because record lengths differ slightly: the short baseline `normal_0hp_97.mat` yields 29 segments, the longest record `IR007_3hp_108.mat` (122917 samples) yields 60, and the remaining 38 yield 59 each.

I treat the acceleration as being in g by convention; the source does not label the unit. Every result here rests on frequencies and on ratios, and neither is affected by that.

2.2 Fault frequencies

The multipliers come from the CWRU specification for the 6205-2RS JEM, and are multiplied by the shaft rate $n = \text{rpm} / 60$.

Table 2. Fault frequency multipliers.

	multiplier	Hz at 1772 rpm ($n = 29.53$ Hz)
BPFO, outer race	3.5848	105.9
BPFI, inner race	5.4152	159.9
BSF, ball	4.7135	139.2
FTF, cage	0.39828	11.8

There is one piece of arithmetic the reader has to follow. At 1 hp, with the shaft at 1772 rpm, $n = 1772 / 60 = 29.53$ Hz, and the outer-race fault frequency is

$$\text{BPFO} = 3.5848 \times 29.53 \text{ Hz} = 105.9 \text{ Hz}$$

The other two follow in the same way. This number is fixed before any measurement is taken, and every criterion later in this report stands on it.

2.3 Three defects I found in the data

I widened `src/verify_data.py` from four hardcoded files to the whole catalogue, and it turned up three problems. All three are now assertions.

(a) The baselines are sampled at 48 kHz, which the source documentation does not state. The apparatus page gives 12 kHz generally and notes 48 kHz for drive-end bearing faults, saying nothing about the baselines. I established it from the data instead. Three machine lines at 145.7 / 159.5 / 189.1 Hz appear in the known-12 kHz fault records; read a baseline as 48 kHz and those lines land at the same frequencies, read it as 12 kHz and they land at a quarter of them. The durations agree too: 10.08 s read as 48 kHz, against the fault records' ~ 10.2 s, where 12 kHz would give 40.3 s. Reading a baseline at 12 kHz puts its entire frequency axis out by a factor of four, and **raises no error at all**. The benchmark study later confirmed this: its Table A1 caption states 48 kHz for the normal baseline data, and its section 3.2 groups the dataset by sampling rate ([3], pp. 126 and 103). I had established it from the data before the paper was available, so I keep the derivation here and cite the paper as confirmation.

(b) File 99 contains a byte-for-byte copy of file 98's channel. It holds `X099_DE_time` and also `X098_DE_time`, identical to file 98's to the last sample. Selecting the first key alphabetically returns file 98's samples under file 99's name — the same data in the dataset twice under two labels, which would leak across any train/test split. My loader now prefers the variable carrying the file's own id. As a matter of fact: the benchmark study's Table 3, its list of records affected by data acquisition problems (p. 106), covers corrupted, clipped and duplicated records, and neither 98 nor 99 appears in it. I state that and nothing stronger.

(c) File 97 is 5.08 s where every other record is about 10 s. The sampling rate is not the problem: six machine lines land where a known 12 kHz record puts them only under a 48 kHz reading, and nowhere near it at 24 or 12 kHz. It is simply a shorter recording. I recorded it as a known exception rather than widening the duration check. The pages of the paper I read do not address it.

Separately, two of the four baselines carry an RPM variable and two do not, and none carries the base-plate channel the fault records have.

I put these three in the data section rather than in the limitations, because finding them was part of the work, and because they are the evidence that this data was checked rather than consumed.

3. Method

3.1 Why the fault frequency is where it is

Each time a rolling element passes over a defect in a raceway it produces an impact. When the defect is stationary, those impacts repeat at a rate fixed jointly by the bearing geometry and the shaft speed. **The identity of the fault is therefore carried by how often the impact recurs**, at the frequency section 2.2 has already given. The diagnostic task is to measure that recurrence rate.

3.2 Why a direct spectrum does not find it

An impact lasting under a millisecond is broadband. Its energy does not go to the frequency at which it recurs; it goes to the structural resonance it excites.

The energy distribution shows this plainly. In the 1 hp, 0.007 in records, **97.4%** of the outer-race record's spectral energy lies in 2000–4000 Hz and only **0.2%** below 500 Hz — which is precisely the band where theory puts the fault frequency. Relative to the healthy record, the outer-race record carries **1382 times** the energy in 3000–4000 Hz and

379 times in 2000–3000 Hz, but **0.5 times** in 0–500 Hz, less than the healthy record has. Taking the Fourier transform of the raw signal therefore means looking for a peak in a band where almost no energy is present.

3.3 Envelope analysis

The remedy has two steps: restrict the signal to the band where the resonance lives, then measure the rhythm of the energy fluctuation within it. In practice that is a fourth-order Butterworth band-pass applied with zero phase, the Hilbert envelope, removal of the mean, and a transform.

The Hilbert transform constructs the analytic signal from a real one; its modulus is the instantaneous amplitude, that is, the envelope. The band-pass has to come first, because the envelope reflects the amplitude fluctuation of whatever is fed to it: without filtering, what comes out is the combined fluctuation of all the machine's activity rather than the part of the resonance that the impacts modulate.

Removing the envelope's mean is not optional. The envelope is a modulus and is strictly positive, so leaving the mean in puts a peak at 0 Hz. I measured that DC peak at **2.3 times** the BPFO peak — not the catastrophe usually described, but enough to halve every real peak on an autoscaled axis and to defeat any automatic peak search.

A convention that holds throughout. Every calculation that compares spectra across records is done on a **common 120976-sample window**: 10.081 s, frequency resolution 0.0992 Hz. That length is the shortest of the four 1 hp records used in those sections, the healthy one. The common length is necessary because frequency resolution is f_s/N , so unequal N would mean unequal line spacing and the comparison would lose its meaning. Per-record time-domain statistics (section 4.1) and the classification features (section 4.6) need no common length and are computed on each record's own.

I verified the chain on a synthetic signal: impacts every 1/105.9 s exciting a 3 kHz decaying resonance, buried under three times as much noise — the physical model above, so the answer is known by construction. A direct FFT scored a peak ratio of **3.0** at 105.9 Hz; the envelope spectrum scored **31.8**, with the peak at **105.90 Hz**.

3.4 Choosing the resonance band

This subsection carries the most weight in the report.

The circularity. The obvious way to replace a band chosen by eye is to score each candidate band by the peak at the fault frequency and keep the best. But that requires knowing which frequency to measure at, and that is exactly what is to be diagnosed. Choosing a parameter with the answer and then deriving the answer from that parameter proves nothing.

What is genuinely known beforehand. The bearing type, its four frequency multipliers and the shaft speed are all known before any diagnosis, so the numerical values of BPFO, BPFI and BSF are known too. Exactly one thing is not: which of the three is faulted. So the legitimate procedure is to score **every** candidate band on **all three** candidate frequencies, keep the highest of the three as that band's score, and record which candidate won. Selecting the highest-scoring band then selects the diagnosis it supports, in the same operation, without using the answer anywhere.

The score: a harmonic comb. The score is the geometric mean of the peak-to-background ratios at $1\times$ to $4\times$ the candidate frequency. A geometric mean requires every harmonic to be present: if any one collapses, the product is dragged down with it. An arithmetic mean would let a single tall line carry the score — and section 3.2 has already shown that a single tall line can be a machine line.

The search grid. 55 candidate bands, widths 500 / 1000 / 2000 Hz, centres every 250 Hz from 250 to 5800 Hz.

Why not kurtosis. Kurtosis is the criterion the kurtogram [2] uses. On this data, the fraction of the best achievable peak ratio that each criterion's chosen band retains is:

Table 3. What each band-selection criterion retains of the best achievable peak ratio.

	kurtosis	comb score
Outer race	23.9%	98.5%
Inner race	64.4%	95.8%
Ball	26.9%	33.8%

Kurtosis measures impulsiveness, and impulses arise from sensor sparks, single mechanical knocks and filter ringing as readily as from bearings. The comb score additionally requires periodicity, which a single event cannot counterfeit. A concrete instance: before I added edge trimming, the kurtosis criterion selected 250–750 Hz for the outer race; trimming 5 percent from each end moved the pick to 4750–5750 Hz. The original selection had been *filtfilt*'s own start-up transient, which raises no error and looks like a result.

The threshold. The highest comb score the healthy record reaches over the **same 55-band search** is **11.1**. Its median over single bands is **3.57** and its 90th percentile **8.55**. Taking a single-band figure as the threshold would ignore that the maximum of many noisy numbers exceeds any one of them, and would manufacture false positives. The threshold has to be calibrated by a procedure carrying the same selection bias.

3.5 Classification

There are nine features in two families. Three are time-domain: RMS, kurtosis, crest factor. Six are envelope: the peak-to-background ratios at BPFO, BPFI and BSF and at their second harmonics.

The envelope features are peak ratios rather than energy shares. Across unseen fault diameters, ratios score **50.7%** against shares' **42.1%**. The difference is in the denominator: a share's denominator is the segment's own energy, which mostly encodes load and fault size — exactly what a classifier should not depend on — whereas a peak ratio's denominator is the local background around that frequency, so overall vibration level cancels.

Segments do not overlap. With a sliding window, adjacent segments share samples, and putting one in training and the other in test is a leak.

The four splits each ask a different question: **A, random shuffle** (which asks almost nothing); **B, grouped by file** (will it recognise a new recording); **C, leave one load out** (will it recognise a new operating condition); **D, leave one fault size out** (will it recognise a fault it has never seen). Each is closer to field conditions than the last.

4. Results

Figures are named `figures/figXXy_*.png` in the repository, numbered by the stage that produced them — Figure 1 here is `figures/fig01_waveforms.png`. Figure numbers in this report run in reading order and so do not track the stage numbers in the filenames.

4.0 An external check on the pipeline

Before any result, the question of whether this pipeline can be trusted. Table B2 of the benchmark study [3] publishes a kurtosis for every record. I computed the same quantity independently, before the paper was available, and the person transcribing the table did not know my values.

All 36 drive-end fault records agree. The median difference is **0.06%**, the largest is **0.3%**, and none exceeds 0.5%. Record 198 reads 2.94 on both sides, record 186 reads 8.84 on both, record 223 reads 9.41 on both, and record 170 is 22.08 here against 22.1 as printed.

The check runs both ways. It confirms the transcription — thirty-six values do not agree by accident — and it validates my loading, channel selection and kurtosis convention against an external published source. **The pipeline is not merely self-consistent; it is checked against the literature.**

The four baselines differ by 0.8 to 3.1 percent, and are expected to: the paper computes on the native 48 kHz signal and I compute on the 12 kHz decimation. The direction and size of that difference are consistent with the resampling, not with an error.

4.1 Time domain

Over the first 0.2 s of the 1 hp, 0.007 in records, the peak accelerations are 0.21 g healthy, 2.85 g outer race, 1.53 g inner race and 0.44 g ball. In that window the outer-race record shows about 21 impulse bursts, and $\text{BPFO} \times 0.2 \text{ s} = 21.2$. **The rhythm is countable by eye in one of the four records and in neither of the other two faults.**

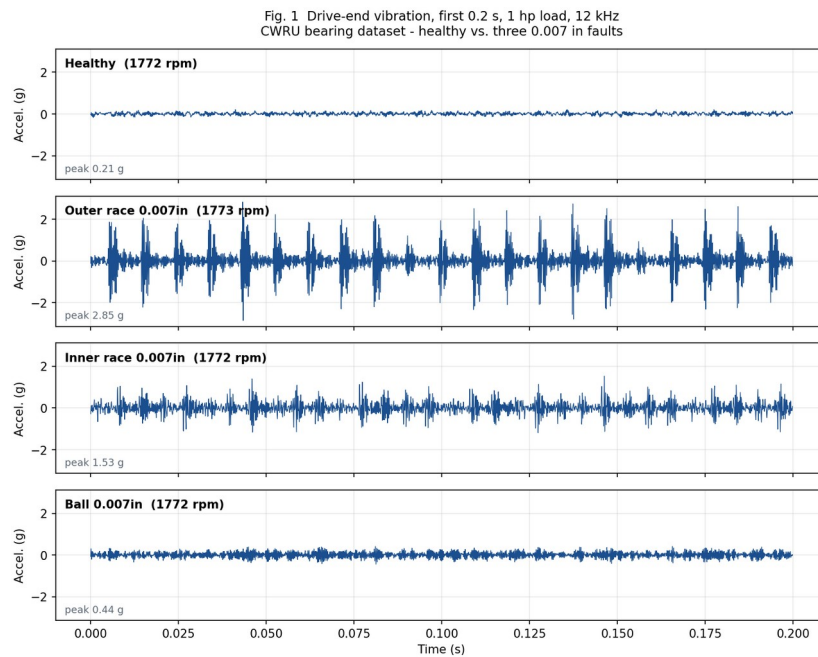


Figure 1. The outer-race fault's impact rhythm can be counted directly off the waveform — about 21 bursts in 0.2 s, against $\text{BPFO} \times 0.2 \text{ s} = 21.2$ — while neither of the other two faults shows a countable rhythm at all.

Table 4. Whole-record statistics, 1 hp, 0.007 in. Kurtosis is in the Pearson convention, where Gaussian noise gives 3.

	RMS (g)	kurtosis	crest factor
Healthy	0.0617	2.98	4.58
Outer race	0.5919	7.59	5.26
Inner race	0.2929	5.54	5.40
Ball	0.1391	2.96	4.74

The healthy record's 2.98 against the Gaussian 3.00 confirms that a sound bearing's vibration carries no impulsive structure. Over 2048-sample segments (59 per record here), kurtosis ranges 2.81–3.37 for the healthy record and 2.67–3.40 for the ball fault: **completely overlapping**. Two of the three statistics therefore miss the ball fault entirely. Only RMS separates it, and RMS needs a healthy baseline from the same machine to mean anything.

Fig. 2 Time-domain statistics, 2048-sample segments (59 per record)
bars = mean, dots = individual segments

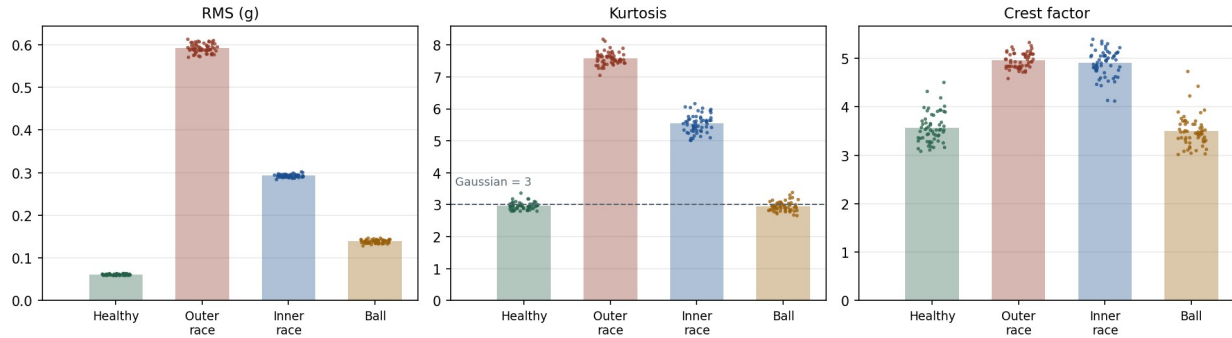


Figure 2. The per-segment statistics of the healthy record and the ball fault overlap completely: two of the three time-domain statistics do not respond to the ball fault at all.

Across fault size, the whole-record kurtosis in descending order is 22.08 (inner 0.014), 21.97 (outer 0.021), 9.41 (ball 0.021), 8.84 (ball 0.014), 7.67 (inner 0.021), 7.59 (outer 0.007), 5.54 (inner 0.007), **2.98 (healthy)**, 2.96 (ball 0.007), 2.94 (outer 0.014). Outer race spans 2.94 to 21.97, inner race 5.54 to 22.08, ball 2.96 to 9.41. **The classes interleave and the healthy record sits inside the range.** The outer-race 0.014 in record reads 2.94 against the healthy 2.98 — a real fault that looks healthy on two statistics of three.

That reading is not an oddity peculiar to this data; it is a documented property of the rig. The same record — data set 198 — has a kurtosis of 2.94 in the benchmark study as well. The paper attributes anomalies of this kind not to fault size or to speed and load but to the rig's assembly: the diagnosis outcomes appear to depend less on fault size or operating condition than on the assembly, which was presumably unchanged within a fault size and changed when a new bearing was installed, the suspected mechanism being mechanical looseness whose severity varied with each installation ([3], p. 107, repeated at pp. 110 and 124). So the reading is explicable, and I reproduced the number to the same decimal without knowing it.

Time-domain statistics can detect. They cannot name.

4.2 The raw spectrum

Computed on the common 120976-sample window, Hann window, resolution 0.0992 Hz.

Table 5. Share of spectral energy per band (%), 1 hp, 0.007 in.

	0–500 Hz	2000–4000 Hz
Healthy	29.4	12.1
Outer race	0.2	97.4
Inner race	0.6	78.1
Ball	1.9	92.7

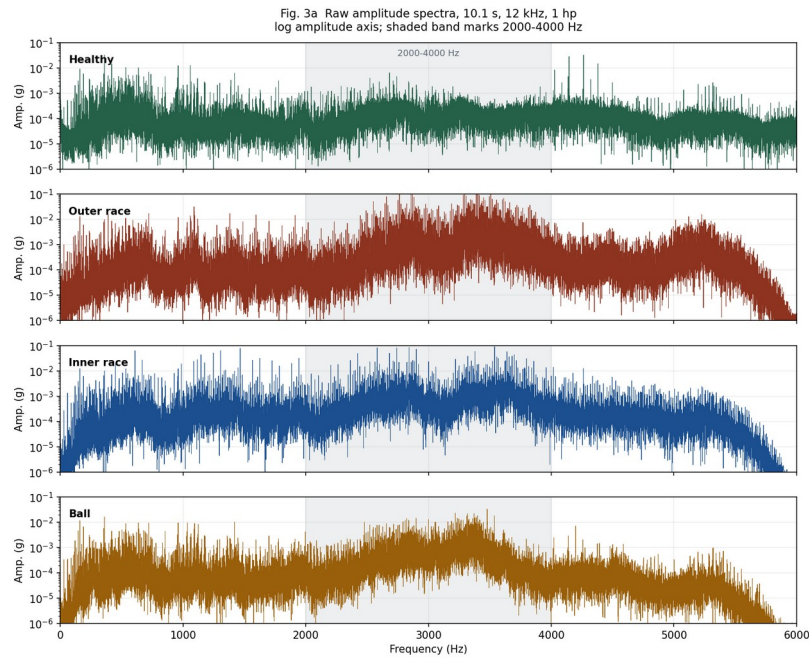


Figure 3. The fault records' energy sits on the structural resonance at 2000–4000 Hz and not in the low band where the theoretical fault frequency lies; the outer-race record holds only 0.2% of its energy below 500 Hz, less than the healthy record does.

Table 6. Amplitude at the theoretical fault frequencies (g).

	BPFO 105.9	BPFI 159.9	BSF 139.2	record RMS
Healthy	0.00062	0.0093	0.00010	0.0617
Outer race	0.00207	0.0131	0.00016	0.5921
Inner race	0.00057	0.0124	0.00066	0.2928
Ball	0.00051	0.0117	0.00052	0.1391

At BPFI all four records carry a comparable peak, the healthy one included, because the 159.5 Hz machine line happens to fall beside BPFI at 159.9 Hz. **Reading the inner-race spectrum alone would confirm an inner-race fault, and the same reasoning applied to the healthy record would confirm one there too.** At BPFO the outer-race record does carry 3.3 times the healthy amplitude, but 0.00207 g against a record RMS of 0.5921 g is **0.35%** — an order of magnitude below the machine lines beside it, and invisible on a linear axis.

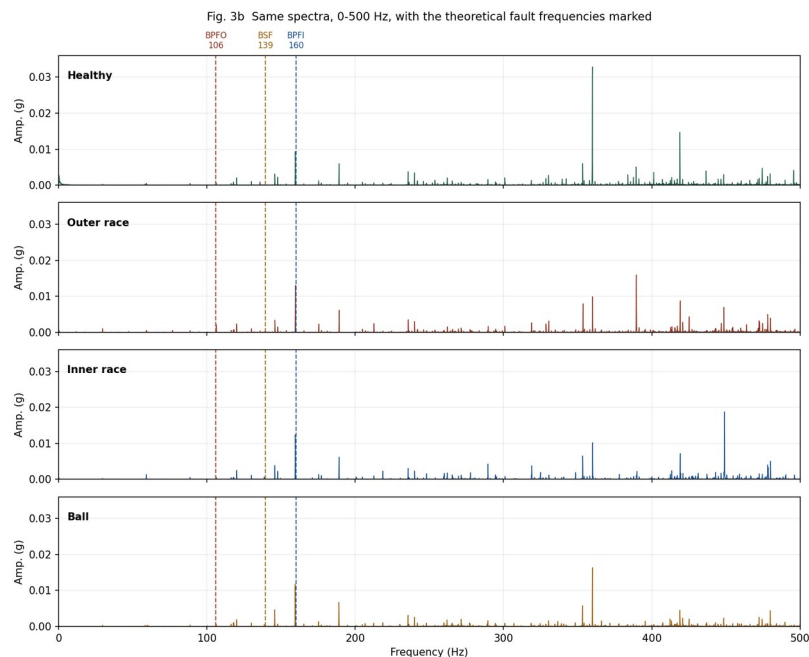


Figure 4. All four records carry a comparable peak at BPFI, the healthy one included, because the 159.5 Hz machine line falls beside it — so reading the low-frequency spectrum alone would diagnose an inner-race fault in a sound bearing.

The strongest lines below 500 Hz are common to all four records: 360.0, 419.0, 159.5, 353.4 and 189.1 Hz. The 360 Hz line is the sixth harmonic of the 60 Hz supply.

4.3 Time–frequency

The STFT uses a 64-sample Hann window and hop 8, giving a 1500 Hz frame rate and 187.5 Hz frequency resolution. **These parameters cannot be copied from a tutorial.** The common 256-sample window at 50 percent overlap leaves a hop of 128 and a frame rate of 93.8 Hz, whose Nyquist limit is 46.9 Hz; BPFO at 105.9 Hz is then not merely blurred but **absent from the frequency axis**. My first attempt raised an empty-slice error rather than returning a low score. The window and hop have to be derived from the rhythm being looked for.

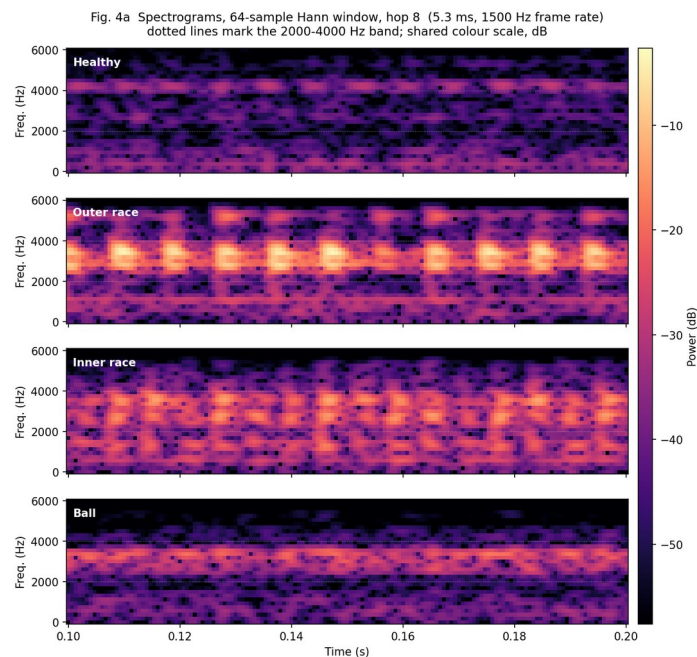


Figure 5. The outer-race fault shows periodic striping in the high-frequency band, so the rhythm is directly visible in the time–frequency plane; the ball fault's band is bright but does not flicker, which is amplitude without periodicity.

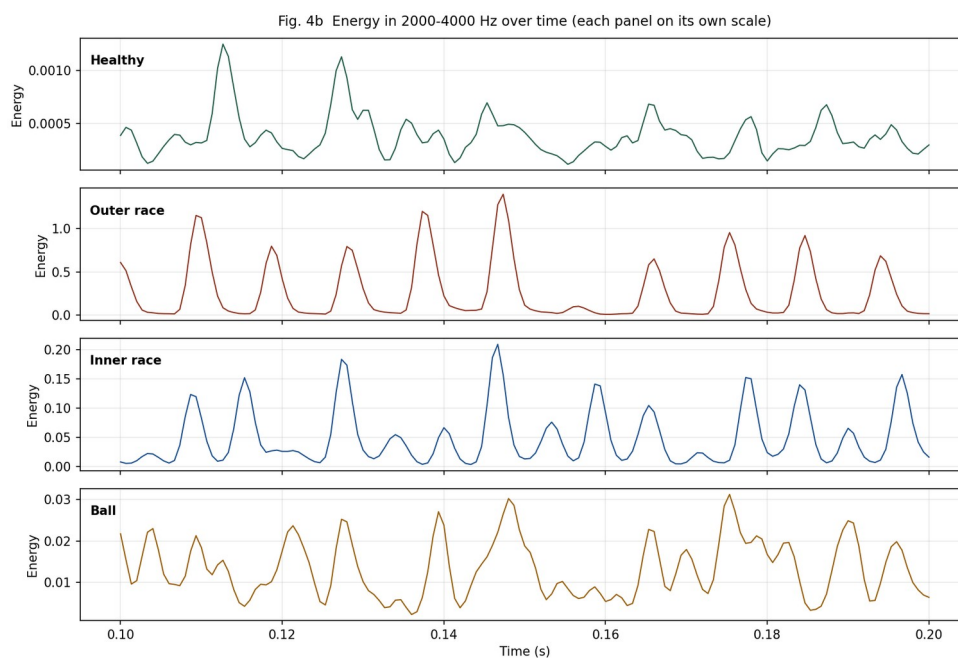


Figure 6. The 2–4 kHz band energy against time is itself the rhythm being sought — the envelope in crude form, with the information in the amplitude modulation rather than in the carrier.

Table 7. Peak-to-background ratio in the spectrum of the 2000–4000 Hz band energy.

record	BPFO	BPFI	BSF
Healthy	3.6	7.2	2.3
Outer race	601.0	9.5	30.2
Inner race	3.3	221.8	3.6
Ball	10.7	10.5	2.3

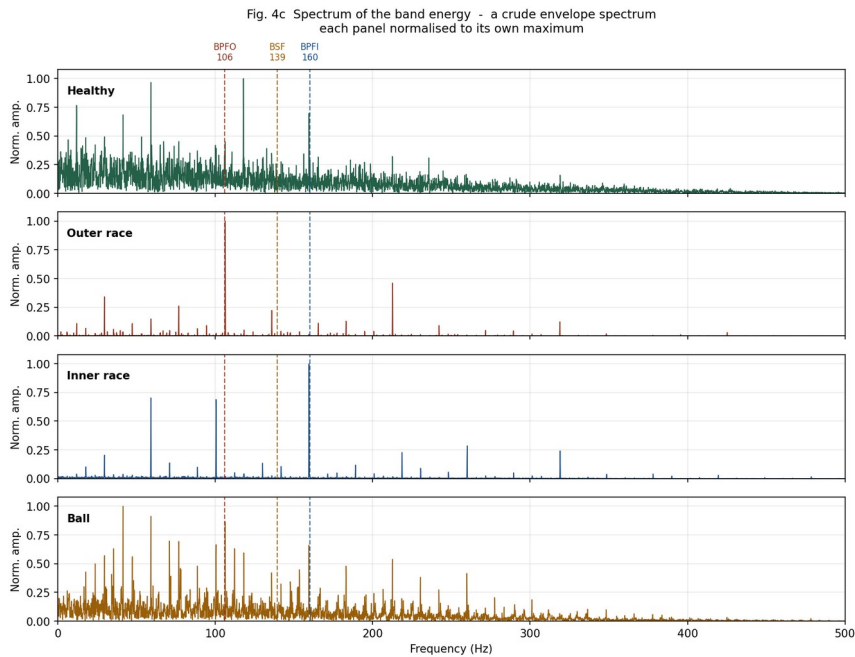


Figure 7. Each record peaks on its own fault frequency and nowhere else — 601 at BPFO for the outer race, 222 at BPFI for the inner race — while the healthy record's highest value of 7.2 sets the false-alarm floor.

The outer race's 1× to 4× BPFO harmonics are 601, 504, 371 and 325. The spectrogram also settles something the numbers alone do not: the ball fault's 3000–3800 Hz band is clearly brighter than the healthy record's, **but it does not flicker**. Raised amplitude is what detection needs; periodicity is what diagnosis needs, and this fault has only the first.

4.4 The envelope spectrum

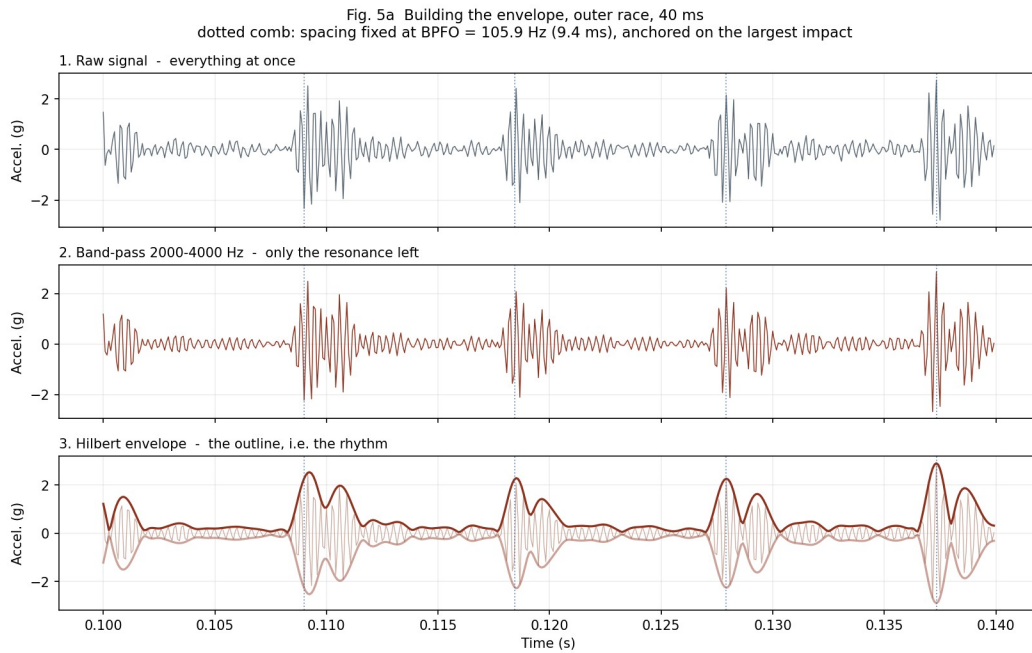


Figure 8. The band-pass confines the signal to the resonance band, and only after the envelope is taken does the rhythm appear — the envelope measures the amplitude fluctuation, independently of the carrier.

Table 8. Envelope-spectrum results, 1 hp, 0.007 in, band-pass 2000–4000 Hz.

record	frequency	ratio	measured	error
Outer race	BPFO 105.9	635	106.3 Hz	+0.4%
Inner race	BPFI 159.9	237	159.5 Hz	−0.25%
Ball	BSF 139.2	2.1	—	—
Healthy	(floor)	4.3 / 9.7 / 2.2	—	—

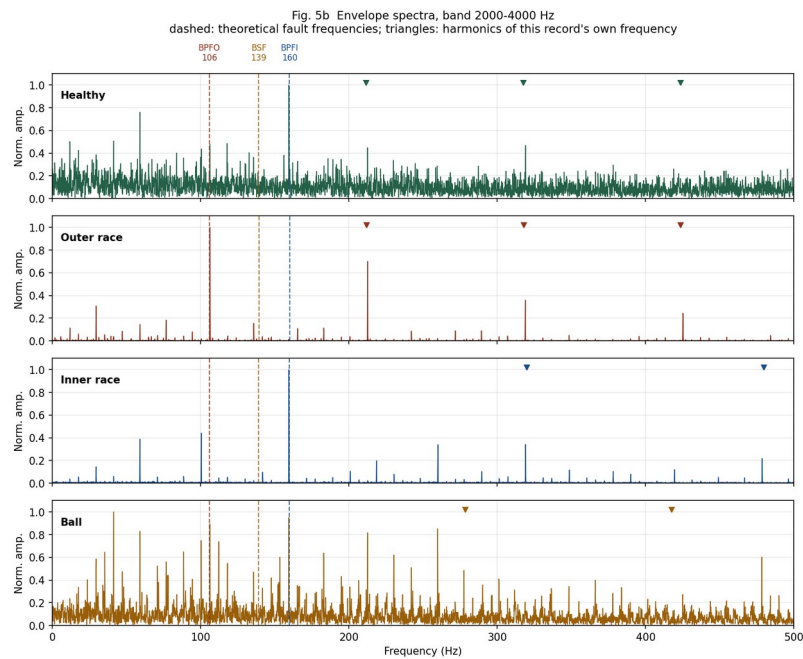


Figure 9. The outer-race and inner-race records each peak only at their own predicted frequency, at ratios of 635 and 237, while the healthy record's tallest line reaches 9.7.

The outer race's $1\times$ to $5\times$ BPFO harmonics are all strongly present. The inner-race record scores 2.9 at BPFO and 4.1 at BSF against 237 at BPFI. Set against the raw spectrum: at BPFI the healthy and inner-race records were within about 33 percent of each other, and after band-passing they differ by a factor of 24, because the 159.5 Hz machine line is removed before the envelope is taken.

A numerical coincidence worth stating explicitly. The inner-race envelope peak is measured at 159.5 Hz, the same figure as the machine line identified in section 4.2. They are not the same object: the machine line is removed by the 2000–4000 Hz band-pass before the envelope is taken, and the 159.5 Hz in the envelope spectrum is a modulation rate, not a line in the original signal. I state this so the coincidence is not read as a confusion.

Two measurements that do not match the textbook. Sidebands spaced at shaft rate are expected around BPFI, because an inner-race fault rotates through the load zone, and not around BPFO, because an outer-race fault is stationary. At the inner race, the ± 1 shaft-rate pair is weak (8.8 and 12.7) while the ± 2 pair is strong (98.7 and 50.8). At the outer race, shaft-rate sidebands are present that are not supposed to be there (106.1 and 113.4). I record both as unexplained observations and do not speculate.

4.5 Selecting the resonance band

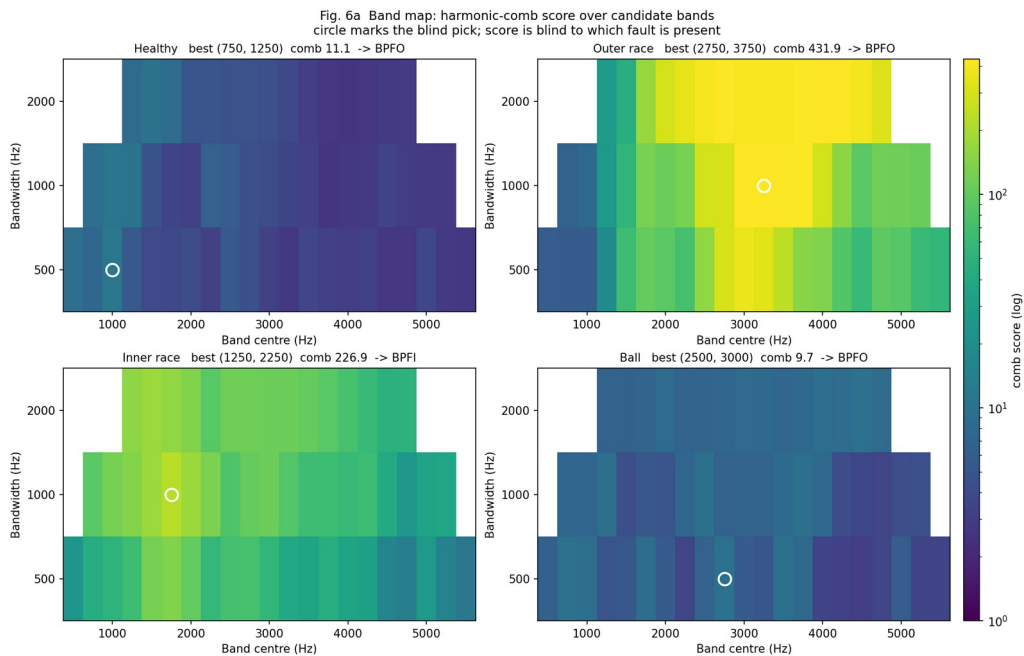


Figure 10. For the outer-race record the high-scoring region at 2500–4000 Hz is a connected area rather than an isolated bright point, so the selection does not depend on precise tuning; the healthy record's map is dark throughout.

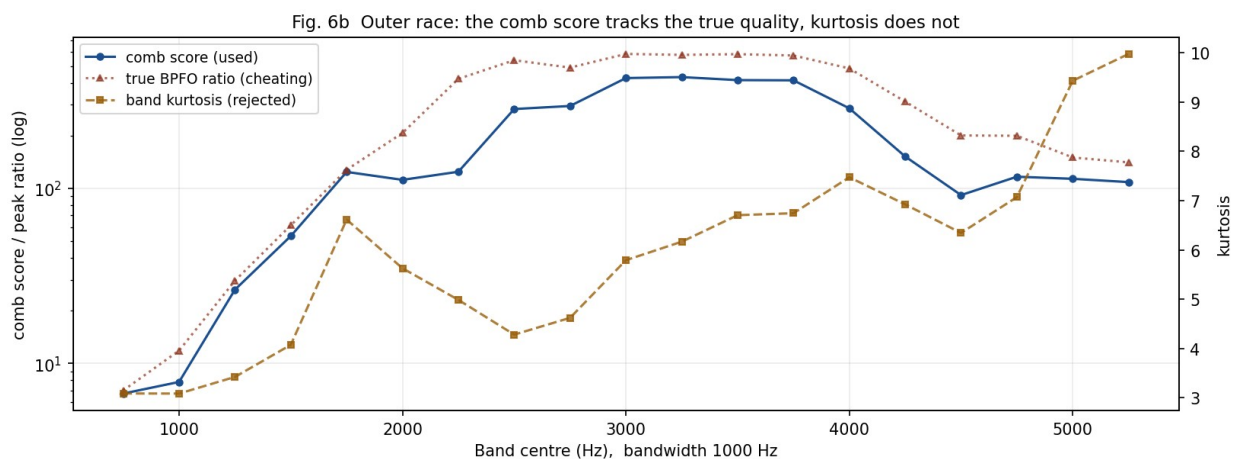


Figure 11. The comb score tracks the best achievable peak ratio almost exactly while kurtosis peaks somewhere else entirely — this figure is the whole argument for not using kurtosis.

Table 9. Verdicts, 1 hp, 0.007 in, threshold 11.06.

record	band (Hz)	comb	× threshold	call	truth
Healthy	750–1250	11.06	1.0	no call	healthy
Outer race	2750–3750	431.90	39.1	BPFO	BPFO
Inner race	1250–2250	226.92	20.5	BPFI	BPFI
Ball	2500–3000	9.66	0.9	no call	BSF

Table 10. Across fault size, 1 hp.

record	band (Hz)	comb	× thr.	call	truth	result
outer 0.007	2750–3750	431.90	39.1	BPFO	BPFO	correct
outer 0.014	750–1750	12.86	1.2	BPFO	BPFO	correct
outer 0.021	1250–2250	85.10	7.7	BPFO	BPFO	correct
inner 0.007	1250–2250	226.92	20.5	BPFI	BPFI	correct
inner 0.014	2500–3000	36.42	3.3	BPFI	BPFI	correct
inner 0.021	2000–4000	69.37	6.3	BPFI	BPFI	correct
ball 0.007	2500–3000	9.66	0.9	—	BSF	declined
ball 0.014	750–1250	10.37	0.9	—	BSF	declined
ball 0.021	3250–4250	14.52	1.3	BPFI	BSF	false

Fig. 6c Robustness across fault size, all at 1 hp

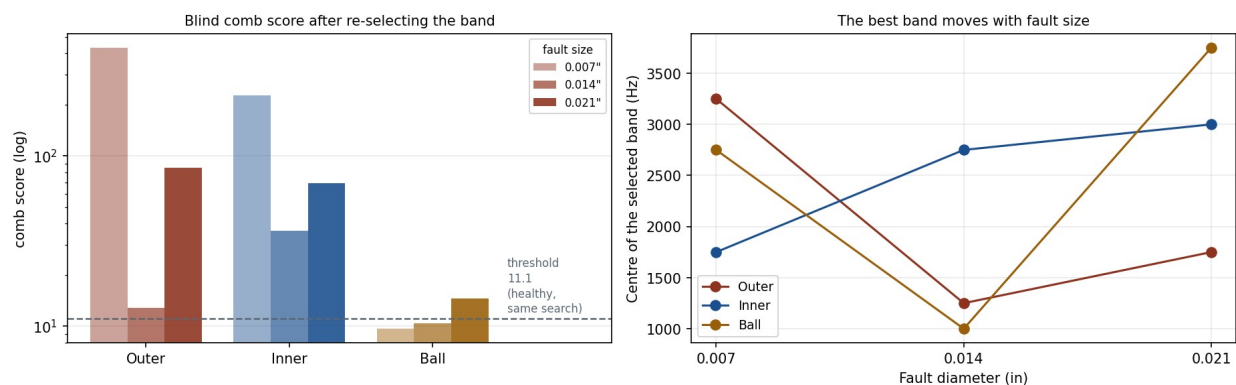


Figure 12. The selected band moves with fault size, but the diagnosis stays correct on all six race records across all three diameters — the method depends on a physical structure, not on a tuned parameter.

All six race faults are diagnosed correctly, even though the selected band moves with both fault location and fault size.

One false positive occurred in nine records: the 0.021 in ball fault scored 1.3 times the threshold, crossed the decision line, and was called an inner-race fault.

It cannot be removed by raising the threshold. The only barely-correct record, outer 0.014 in at 1.2×, and the false one, ball 0.021 in at 1.3×, are not separable by this score. So I replaced the single threshold with a three-tier rule:

Table 11. The three-tier decision rule.

comb / threshold	verdict	on these records
> 5×	diagnosis	4 records, 4 correct
1× to 5×	tentative, re-measure	3 records, 2 correct, 1 wrong
< 1×	no call	2 records

Every error falls in the tier that admits to being unsure.

4.5.1 How unstable the grey tier is

The three-tier rule is not a patch invented afterwards to explain that one false positive. Scoring the same nine records on their own full lengths instead of the common window — a difference of at most about 2000 samples, under 2 percent — gives:

Table 12. Effect of record length on the verdict.

record	common window (120976)	full record	
outer 0.007	431.90 correct	462.28 correct	
outer 0.014	12.86 correct	13.52 correct	
outer 0.021	85.10 correct	87.35 correct	
inner 0.007	226.92 correct	224.33 correct	
inner 0.014	36.42 correct	36.77 correct	
inner 0.021	69.37 correct	70.51 correct	
ball 0.007	9.66 declined	9.76 declined	
ball 0.014	10.37 declined	11.28 false	verdict flips
ball 0.021	14.52 false	14.29 false	

Exactly one record changes verdict, and it is the one nearest the 11.06 threshold. Every record more than a few percent clear of the threshold keeps its verdict. This is the strongest evidence for the three-tier rule: inside the 1×-to-5× tier a verdict does not survive a two percent change in how much data is used, so a single number there cannot be treated as a diagnosis. Above 5×, nothing moves.

4.5.2 Outer race 0.014 in: two marginal assessments that agree

The one barely-correct record in Table 10 is outer race 0.014 in, data set 198. The benchmark study's assessment of that group is marginal too. Discussing outer-race faults centred in the load zone, the paper states that the next four series of data could not be diagnosed by any of the techniques applied ([3], p. 110), which in that section are sets 197 to 200 — exactly this project's 0.014 in outer-race group. Its Table B2 grades 198 on the drive-end channel as P2 under Method 1, meaning potentially diagnosable, with smeared components appearing to coincide with the expected frequencies, and N2 under Methods 2 and 3, meaning not diagnosable and virtually indistinguishable from noise.

The correct statement is therefore that two independent marginal assessments agree. The benchmark grades this record potentially diagnosable at best; the rule proposed here returns the correct fault location, but at 1.2× its threshold, inside the tier that section 4.5.1 shows to be unstable against a two percent change in record length. Neither is a confident diagnosis, and they agree about that.

This result **is not evidence of improvement over published methods**, and I make no such claim. It is a correct call inside an interval that my own method marks as tentative, and nothing more.

One inconsistency inside the paper, which I record rather than resolve: the text on p. 110 says data set 197 is the only 0.014 in case with a partial diagnosis, while Table B2 also shows P2 for 198 and P1 for 199 on the drive-end channel under Method 1.

4.5.3 If one fixed band must be used

Taking the geometric mean of the peak ratio over the six race records, the candidate fixed bands score: 1000–3000 Hz, **110**; 2000–3000 Hz, **102**; 2000–4000 Hz, **93.6**; 2750–3750 Hz, **80.7**; 1250–2250 Hz, **56.6**. The 1000–3000 Hz

band is best on none of the six individually and poor on none. The 2000–4000 Hz band used throughout section 4.4, which I had chosen by eye from a figure, drops to 5.9 on the outer-race 0.014 in record.

4.6 Classification

40 records, 2048-sample non-overlapping segments, 2331 rows, 9 features, envelope band 1000–3000 Hz, RBF SVM (C = 10) on standardised features.

Table 13. Accuracy by split.

split	time (3)	envelope (6)	all (9)
A random shuffle	97.0%	77.9%	95.9%
B grouped by file	96.3%	76.1%	93.5%
C leave one load out	96.5%	75.6%	93.9%
D leave one fault size out	33.1%	50.7%	48.3%

Fig. 7b The same model and the same data, evaluated four ways

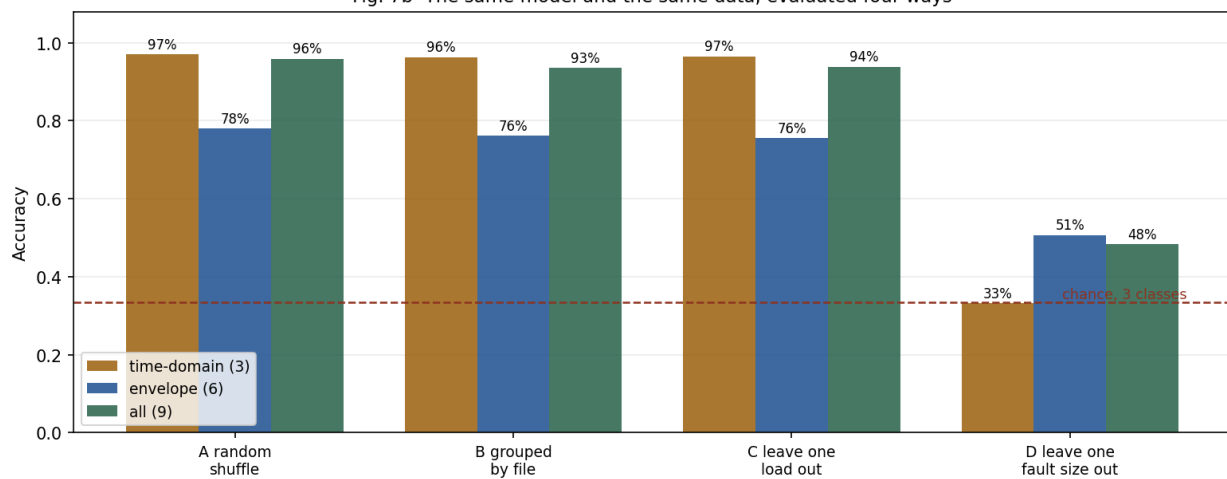


Figure 13. The same model, the same data and the same features give accuracies from 97.0% to 33.1% depending only on how the data is split — so an accuracy quoted without its split cannot be interpreted.

Split D tests the three fault classes only, where chance is 33.3 percent. **The time-domain features score 33.1 percent — not weak, empty.** Grouping by file costs only 2.4 points, because holding out one record leaves the same fault at the same diameter under three other loads in the training set. **The leak that matters is between conditions, not between segments.**

Table 14. Split D, by held-out diameter.

	0.007	0.014	0.021
time (3)	33.3%	29.8%	36.3%
envelope (6)	58.4%	31.6%	62.0%
all (9)	65.4%	30.8%	48.7%

The 0.014 in column is the worst for every feature set. That is the same fact as in section 4.1: the group contains data set 198, whose anomaly comes from the rig's assembly rather than from the fault size ([3], p. 107). The classifier's difficulty on that column is therefore not a failure of the model but a property of a group of records that the published assessment also graded as hard to diagnose.

Fig. 7a Confusion matrices, all nine features, row-normalised

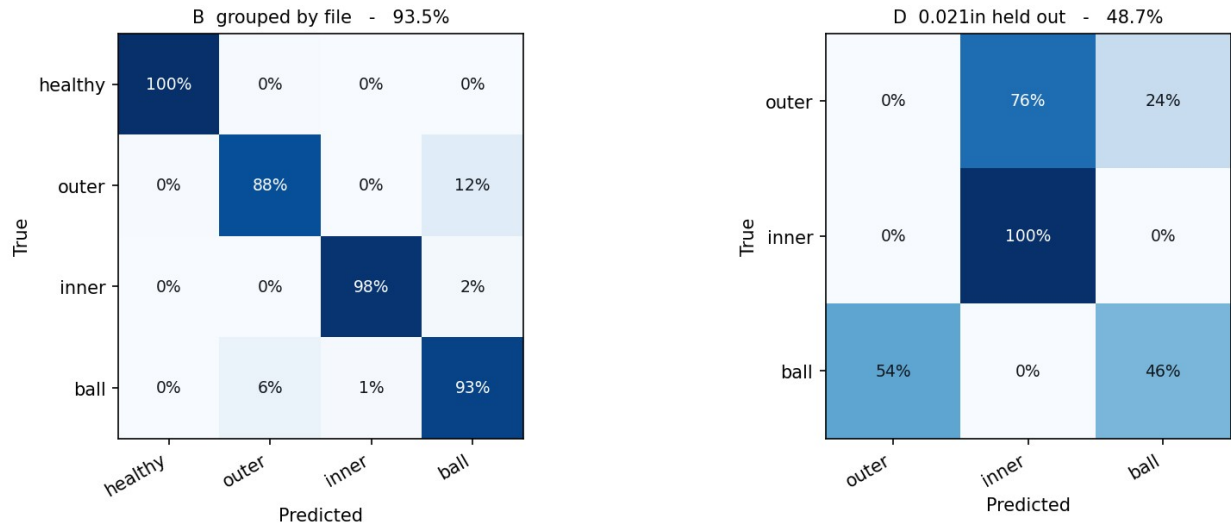


Figure 14. With 0.021 in held out, the overall 48.7% is carried entirely by one class: outer-race recall is 0%, and the model identifies not one outer-race segment.

With 0.021 in held out and all nine features, overall accuracy is 48.7 percent, and the per-class recalls are: outer **0%**, inner 100%, ball 46.2%. **Accuracy alone conceals a model that identifies not one outer-race segment.** Under split B the overall figure is 93.5 percent, with the main confusion between outer race and ball.

Fig. 7c Colour = fault location, marker = fault diameter
each tight cluster is one record

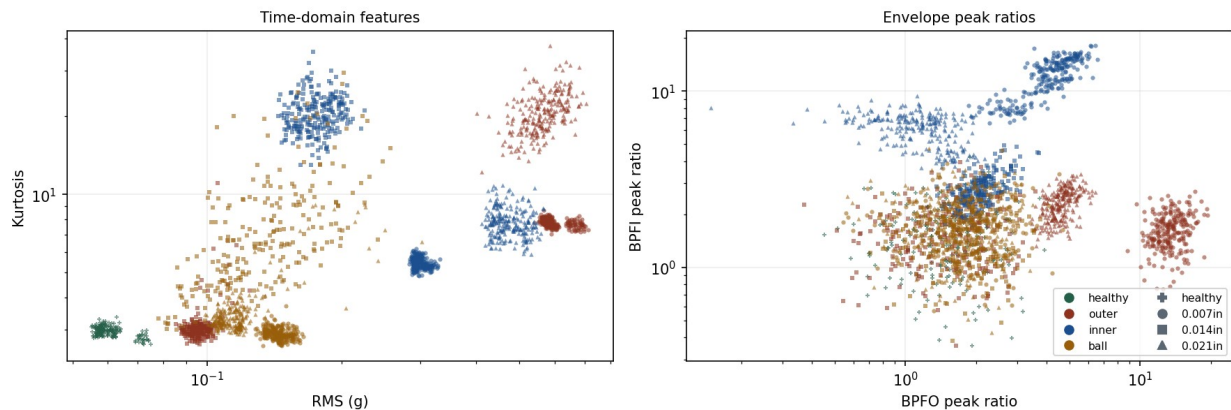


Figure 15. The tight clusters of the time-domain features are arranged by fault diameter rather than by fault class, so holding out a diameter removes all information about that group — this is where the 33.1% comes from.

Table 15. Effect of segment length.

segment length	B grouped by file	D unseen size
512	87.7%	54.7%
1024	91.5%	60.6%
2048	93.5%	48.3%
4096	94.2%	47.3%

The best segment length depends on which split is being optimised. **The 2048 samples I used throughout is best on neither.**

Against the method that does not learn. On the same unseen diameters, the rule of section 3.4 diagnoses all six race records correctly, having never been trained. Learning a map from waveform to label does not extrapolate to a fault that was not in the training set; measuring a frequency that geometry predicts does.

5. Discussion and limitations

5.1 What the results mean

The two methods fail differently, and that difference is this report's conclusion.

The rule of section 3.4 measures a quantity that geometry predicts. Its inputs are the bearing type, the shaft speed and the three candidate frequencies computed from them, all of which exist before any diagnosis and none of which depends on how large the fault is. So when the diameter becomes one the training set never contained, the rule still holds — it never depended on having seen it.

The classifier learns a map from waveform to label. That map changes with fault size, so it fails as soon as it leaves the shapes it has seen. It is worth being precise about why, because the obvious reading is wrong: **the classifier is not short of physical information**. Six of its nine features are the peak ratios at the three geometry-derived fault frequencies and their second harmonics — precisely the quantity sections 3.3 and 3.4 are built on. That information does work: across an unseen diameter the envelope features alone reach 50.7 percent where the time-domain features reach 33.1. But nothing obliges the model to use it, and on the training set identifying a record by its tight cluster is both easier and more accurate than comparing three peak ratios. It took the easier route, and the easier route is the one that does not survive a new fault size.

So the lesson is not to add physics to the features. The physics was in the features. The lesson is that an objective which rewards fitting the training set will prefer whatever fits it, and a feature's physical meaning gives it no standing in that competition.

The time-domain statistics scoring 33.1 percent against a 33.3 percent baseline is **the sharpest single result in this report** — not a weak effect, no effect. It confirms the observation in section 4.1 from the other direction: those three statistics are unchanged when the sample order is shuffled, and the fault type lives only in the order.

5.2 Limitations

Each of these is a finding rather than an apology. Each names a specific place where the boundary of this result lies.

1. One public dataset, one bearing type, one test rig. Laboratory data is far cleaner than a plant floor, and every number here should be read under that condition.

2. The faults are single points cut by EDM, not the spalling that real bearings develop in service. The defect geometry differs, and so does the impact it excites.

3. The ball fault is not diagnosed at any diameter. Its band is energetic but unmodulated — detection without diagnosis. This is the clearest failure in the project, and the benchmark study arrives at the same place by different routes. Its Table 5 lists the records that give classical symptoms under at least one method, and the ball column is **empty for all three groups** — 12 k drive end, 48 k drive end and 12 k fan end. No ball fault anywhere earns a Y1. Its Table 6 lists the records not diagnosable by any applied method, and for 12 k drive-end ball faults it includes **118 and 119; record 119 is this project's 1 hp ball 0.007 in record, the one my rule declined to call**. The paper states that the ball fault cases are the most difficult to diagnose, with only a few showing the classic envelope-spectrum symptoms, and that the only sets diagnosable by direct envelope analysis of the raw signal come from the 0.021 in and 0.028 in faults ([3], p. 108, with the finding repeated for the other two groups at pp. 115 and 119).

This does not make my method better. It makes the failure legible: the obstacle is a property of the data that three established techniques also failed to overcome. The direction of agreement on record 119 is worth noting on its own — my rule declined to diagnose it and the benchmark found it not diagnosable. A method that refuses where the literature also fails is behaving correctly.

4. One false positive in nine, and the threshold cannot separate it from the only marginally correct record. Section 4.5.1 shows further that a verdict in that tier does not survive a two percent change in record length. The three-tier rule handles how the problem presents; it does not remove the problem.

5. The classifier was not tested across speed regimes far from these four, nor on a different machine. Split C shows that cross-load generalisation is not difficult on this data, and that should not be extrapolated to cross-machine generalisation.

6. The comb score's harmonic count and the band grid were fixed by hand. Only a coarse-versus-fine grid check was done; there is no systematic sensitivity analysis.

7. The segment length used throughout is best on neither split: 1024 samples generalises better ($D = 60.6\%$) and 4096 scores higher within a split ($B = 94.2\%$).

5.3 What would come next

Three concrete things. Test the rule of section 3.4 on a second dataset, to separate a method that works from a method that works on this data. Apply cyclostationary methods to the ball fault, since both this report and the benchmark study indicate that its energy is present but lacks the periodic modulation this method depends on. And calibrate the threshold per machine, from a healthy baseline recorded on that machine under the identical search, rather than carrying over the 11.1 obtained here.

6. What I learned

The most valuable material in this project is the mistakes, because each has a general lesson attached.

A figure whose axes were four times wrong and looked fine. The baselines are sampled at 48 kHz and the documentation does not say so. Read at 12 kHz, the code raised no error, the plots looked normal and every line was present — the entire frequency axis was simply displaced by a factor of four.

A duplicate record hidden inside another file. File 99 contains a byte-for-byte copy of file 98's channel; taking keys in alphabetical order reads the same data under two labels and leaks it across a train/test split.

Reaching for a standard criterion and finding it selecting a filter artefact. The band kurtosis chose 250–750 Hz for the outer race, which was `filtfilt`'s start-up transient; trimming 5 percent from each end made that choice disappear.

Aligning a theoretical comb from $t = 0$ and concluding the theory was wrong, when the only thing I could not know was the phase.

Finding that grouping by file, the usual advice, did not close the leak that mattered. It cost only 2.4 accuracy points, because the other records from the same operating condition were still in the training set.

One thread runs through all five: **each was found by comparing against something independent** — a healthy record, a file of known sampling rate, a synthetic signal whose answer was known by construction, a second fault size. **Not one of them raised an error**. The code ran, and the result looked like a result.

Section 4.0 is the last and strongest link in that thread: the kurtosis of 36 records agrees, value by value, with a published paper I had not yet read when I computed them. Before that check I could say only that the pipeline was self-consistent; after it, the pipeline has an external reference.

That is the main judgement I take away from this project. Code that does not raise an error is no evidence at all that it computed the right thing; the only thing that tests it is a comparison against an independent source whose answer you already know.

Appendix

A. Repository

<https://github.com/Mmaotttt/bearings-adventure>

Public, default branch `main`. The records themselves are not in it; they are fetched by `src/download_data.py` and catalogued in `src/dataset.py`, so the repository is about five megabytes and anyone can reproduce every figure from it.

`data/` holds the CWRU records, which are fetched by `src/download_data.py` rather than committed; `src/` holds the loading, envelope-analysis and feature code; `notebooks/` works through the seven stages in order; and `figures/` holds every figure in this report, each regenerated by the notebook that produced it.

B. Data checks — output of `src/verify_data.py`

Run on 2026-09-21. Reproduce with `python src/verify_data.py`. Only the four baseline rows and the closing assertion line are reproduced here: all three defects of section 2.3 live in the baselines, so the evidence is concentrated there and the remaining 36 rows only repeat the confirmation. The full 40-row output is in the repository.

40 of 40 catalogue records on disk

file	variable	s	rpm	BPF0	BPFI	BSF	label
normal_0hp_97.mat	X097_DE_time	5.08	1796	107.3	162.1	141.1	healthy 0hp
normal_1hp_98.mat	X098_DE_time	10.08	1772	105.9	159.9	139.2	healthy 1hp
normal_2hp_99.mat	X099_DE_time	10.11	1750	104.6	157.9	137.5	healthy 2hp
normal_3hp_100.mat	X100_DE_time	10.12	1725	103.1	155.7	135.5	healthy 3hp

... 36 further fault records omitted here; see the repository ...

all 40 records on the 12000 Hz grid, speeds consistent with their load, durations ~10 s, no two holding the same samples. The final line is the assertion set: one sampling grid, speeds consistent with the nominal for each load, durations near ten seconds, and no two records holding identical samples. The last of those exists because of defect (b) in section 2.3. Note also that the `normal_2hp_99.mat` row takes `X099_DE_time` and not `X098_DE_time`, which is the fix for defect (b) shown to be in force.

C. Environment

Python	3.12.1
numpy	2.4.4
scipy	1.17.1
matplotlib	3.10.8
scikit-learn	1.8.0
jupyterlab	4.6.3
platform	Windows 11, AMD64

D. References

Bibliographic details checked against CrossRef on 2026-09-24.

- [1] Randall, R. B., & Antoni, J. (2011). Rolling element bearing diagnostics — a tutorial. *Mechanical Systems and Signal Processing*, **25**(2), 485–520. doi:10.1016/j.ymssp.2010.07.017
- [2] Antoni, J. (2007). Fast computation of the kurtogram for the detection of transient faults. *Mechanical Systems and Signal Processing*, **21**(1), 108–124. doi:10.1016/j.ymssp.2005.12.002

[3] Smith, W. A., & Randall, R. B. (2015). Rolling element bearing diagnostics using the Case Western Reserve University data: a benchmark study. *Mechanical Systems and Signal Processing*, **64–65**, 100–131. doi:10.1016/j.ymssp.2015.04.021

Every reference to [3] in this report is a paraphrase with a page citation. The paper could not be copied as text and was read as page images and transcribed by hand; the kurtosis values of its Table B2 have been confirmed independently by the 36-record check of section 4.0, but its prose sentences have not been confirmed a second time, so I do not place any of its wording inside quotation marks.

The tables used here are the 12 k drive-end ones, A2 (p. 126) and B2 (p. 128); the grading definitions are in its Table 4 (p. 107).